



AUTOMATIC RAILWAY GATE CONTROL SYSTEM

Using IR Sensor Technology

Harini • Prototype Development Review (Phase 4)

Problem & Objective



The Problem

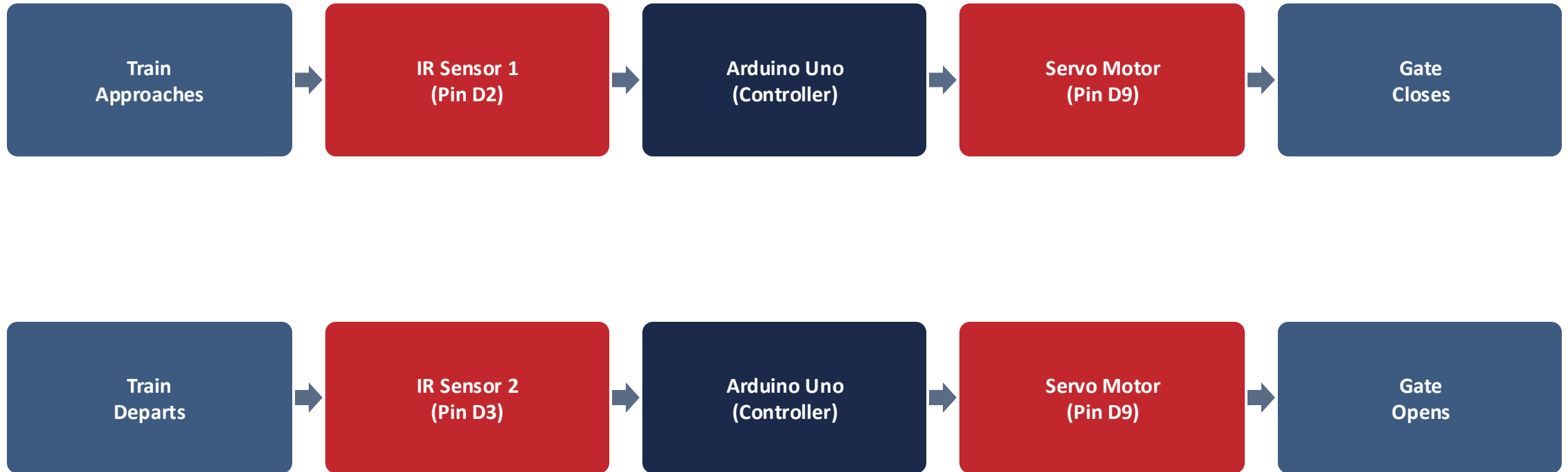
- Manually operated level-crossing gates depend on a human operator's timing and attention.
- Delayed closing risks collisions; delayed opening causes unnecessary traffic hold-ups.
- Remote or unmanned crossings have no reliable automated safeguard.



The Objective

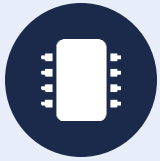
- Automate the gate using low-cost IR sensors that detect an approaching and departing train.
- Drive the barrier with an Arduino Uno and a servo motor — no manual intervention.
- Improve crossing safety and reduce needless wait time for road traffic.

How It Works — System Flow



Gate stays closed for the full duration the train is between the two sensors.

Hardware Components



Arduino Uno

Central controller running the gate logic



2× IR Sensor Modules

Detect train arrival (D2) & departure (D3)



SG90 Micro Servo

Actuates the barrier arm (Pin D9)



Breadboard + Track Model

Clean power distribution & physical demo

Pin Mapping & Wiring

Component	Arduino Pin	Notes
IR Sensor 1	D2 (digital in)	Detects approaching train
IR Sensor 2	D3 (digital in)	Detects departing train
Servo Motor	D9 (PWM out)	Drives the barrier arm
All modules	5V / GND	Shared power rail on breadboard

Safe wiring order: connect GND first → VCC → signal wires → power on last.

IR sensor signal polarity must match the code logic — verify HIGH/LOW behavior before final assembly.

Prototype Status

2

IR sensors
integrated & tested

1

Servo-actuated
gate mechanism

4

Project phases
completed through PDR



Working prototype demonstrated on a physical track model; gate closes on Sensor 1 trigger and reopens on Sensor 2 trigger, matching the design logic.

Future Enhancements



- LED & sound signals at the crossing for added visual and audible warning
- IoT-based remote monitoring of gate status and sensor health
- Redundant sensor pairs to guard against single-sensor failure
- PCB fabrication for a more compact, field-ready control unit

Thank you — questions welcome.